

# Domain Knowledge Document

## Meta Ad Performance Dataset

### 1. About the Data

This dataset represents **Meta Ads Performance Data**, covering campaigns, ads, user demographics, and ad interaction events. It is modelled after how **Facebook/Instagram (Meta)** ad platforms capture data.

The purpose of this dataset is to **analyse advertising performance**, optimize targeting, and measure **ROI (Return on Investment)** through KPIs such as:

- Impressions (how often ads are seen)
- Clicks (engagement with ads)
- Purchases (conversions)
- CPM, CPC, CTR, and ROAS (efficiency metrics)
- Audience insights (demographics, location, interests)

This dataset is ideal for building a **Power BI Dashboard** to evaluate campaign effectiveness, budget utilization, and user engagement patterns.

### 2. Use of Each

#### Table Table 1:

##### ad\_events

- Stores **event-level logs** (like impressions, clicks, purchases).
- This is the **fact table** in the model because all KPIs are derived from events.
- Used to analyze **when and how users interact with**

#### ads. Table 2: ads

- Contains details of each ad creative.
- Defines targeting criteria and which campaign an ad belongs to.
- Used for **platform-level** and **creative-type-level analysis** (e.g., Facebook Video Ads vs Instagram Image Ads).

#### Table 3: campaigns

- High-level campaign strategy and budget allocation.
- Provides **timeframe** and **budget** for ads.
- Used to calculate cost-based KPIs (CPC, CPM, ROAS).

**Table 4: users**

- Stores demographic and interest information of users who interact with ads.
- Used for **audience segmentation** (gender, age, country, location, interests).
- Helps analyze **targeting efficiency** (are ads reaching the right audience?).

**3. Table and Field**

**Details Table 1:**

**ad\_events**

**Purpose:** Captures every interaction (event) between a user and an ad.

| Field       | Description                                                | Example             | Use in Analysis                                    |
|-------------|------------------------------------------------------------|---------------------|----------------------------------------------------|
| event_id    | Unique identifier for each event                           | 100234              | Used as primary key for the table                  |
| ad_id       | Links to ads table                                         | 501                 | Join with ads → get ad_platform, ad_type           |
| user_id     | Links to users table                                       | U_1204              | Join with users → get demographics                 |
| timestamp   | Exact time of event                                        | 2025-03-12 14:30:00 | Build date hierarchy (Day, Week, Month)            |
| day_of_week | Derived field: day of the week                             | Tuesday             | To compare weekday vs weekend performance          |
| time_of_day | Derived field: segment of day                              | Afternoon           | See when users engage most                         |
| event_type  | Type of event: Impression, Click, Share, Comment, Purchase | Click               | Funnel analysis (Impressions → Clicks → Purchases) |

**Usage:** This is the foundation for **KPIs** such as Impressions, Clicks, CTR, Conversion Rate, and ROAS.

**Table 2: ads**

**Purpose:** Defines ad-level metadata.

| Field            | Description                                                               | Example         | Use in Analysis                        |
|------------------|---------------------------------------------------------------------------|-----------------|----------------------------------------|
| ad_id            | Unique ad identifier                                                      | 501             | Primary key; joins to ad_events        |
| campaign_id      | Campaign association                                                      | C_101           | Join to campaigns table                |
| ad_platform      | Platform where ad runs (Facebook, Instagram, Messenger, Audience Network) | Instagram       | Compare platform performance           |
| ad_type          | Creative format (Image, Video, Carousel, Story)                           | Video           | Performance by creative type           |
| target_gender    | Gender targeted                                                           | Female          | Check targeting efficiency             |
| target_age_group | Age group targeted                                                        | 25–34           | Compare target vs actual engagement    |
| target_interests | Topics/interests targeted                                                 | Travel, Fashion | Check match with actual user interests |

**Usage:** Helps identify **which platform + ad type combination works best**, and whether targeting matches actual user engagement.

**Table 3: campaigns**

**Purpose:** Contains campaign-level information (budget, duration, strategy).

| Field       | Description          | Example             | Use in Analysis            |
|-------------|----------------------|---------------------|----------------------------|
| campaign_id | Unique campaign ID   | C_101               | Primary key; joins to ads  |
| name        | Campaign name        | "Spring Promo 2025" | Reporting, filtering       |
| start_date  | Campaign launch date | 2025-03-01          | Track active campaigns     |
| end_date    | Campaign end date    | 2025-03-31          | Campaign duration analysis |

| Field         | Description                   | Example  | Use in Analysis              |
|---------------|-------------------------------|----------|------------------------------|
| duration_days | Derived: campaign length      | 30       | Compare pacing & performance |
| total_budget  | Budget allocated for campaign | \$50,000 | Basis for CPM, CPC, ROAS     |

**Usage:** Enables **budget tracking, pacing, and ROI analysis.**

**Table 4: users**

**Purpose:** Demographic and interest details of users engaging with ads.

| Field       | Description                             | Example      | Use in Analysis                    |
|-------------|-----------------------------------------|--------------|------------------------------------|
| user_id     | Unique user identifier                  | U_1204       | Primary key; joins to ad_events    |
| user_gender | Gender of user                          | Male         | Gender-based performance           |
| user_age    | Age of user                             | 27           | Basis for custom segmentation      |
| age_group   | Grouped age bucket (18–24, 25–34, etc.) | 25–34        | Compare audience engagement by age |
| country     | User’s country                          | India        | Country-level reach analysis       |
| location    | More specific location (city/state)     | Bangalore    | Geo-targeting                      |
| interests   | User’s interests                        | Tech, Travel | Match vs targeting interests       |

**Usage:** Helps measure **audience targeting accuracy** (e.g., Ads targeted at women 18–24 vs actual engagement from men 25–34).

#### 4. How the Tables Work Together

- **ad\_events** → **ads** → Links events to ad details (platform, type, targeting).
- **ads** → **campaigns** → Links ads to campaign metadata (budget, duration).
- **ad\_events** → **users** → Links user engagement events to demographic data.

This creates a **star schema**:

- **Fact Table:** ad\_events
- **Dimension Tables:** ads, campaigns, users